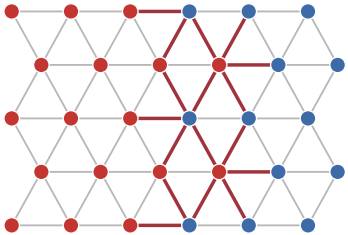
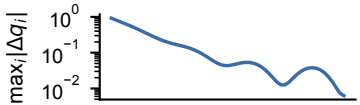
(a) Potts HMRF on the spot graph



$$P(y, z) \propto \exp\left(\sum_i \log p(y_i \mid z_i, \theta) + \beta \sum_{i \sim j} \mathbf{1}[z_i = z_j]\right)$$

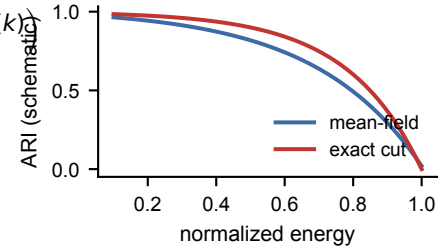
(b) Mean-field VI: fixed-point updates

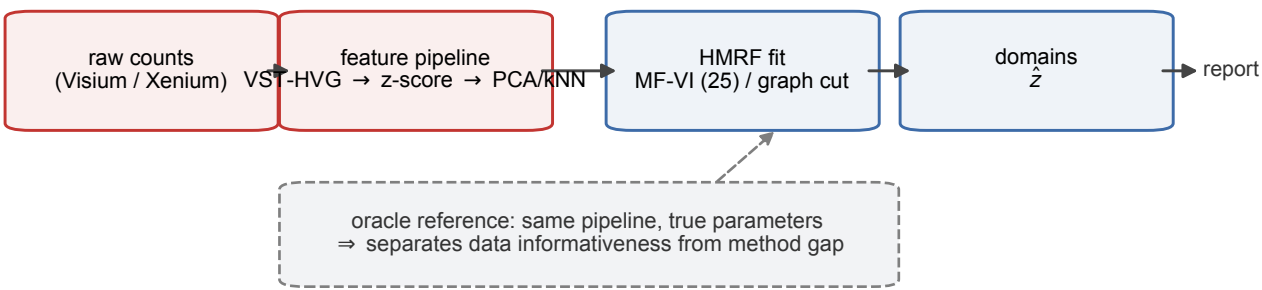
$$q_i(k) \propto \exp\left(\log p(y_i \mid k, \theta) + \beta \sum_{j \in N(i)} q_j(k)\right)$$



25 vectorized restarts
(best / median protocols)

(c) Exact graph cuts (α-expansion)



lower energy ≠ higher label accuracy:
ARI gap ≤ 0.006 with sign flips

(d) End-to-end pipeline and oracle attribution